

Predictive_PS6 & PS7_735

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2026-03-13

PREDICTIVE ANALYTICS

PROBLEM SHEET 6: Model Selection

To demonstrate the application of subset selection methods

Consider the “Hitters” data. For loading the data you need to install ISLR library. We wish to predict a baseball player’s salary on the basis of various predictors. So first we need to choose an optimal model.

```
library(ISLR)
```

```
## Warning: package 'ISLR' was built under R version 4.5.2
```

```
hit = Hitters
```

```
head(hit)
```

```
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits
CHmRun
## -Andy Allanson      293   66     1   30  29    14     1    293    66
1
## -Alan Ashby         315   81     7   24  38    39    14   3449    835
69
## -Alvin Davis        479  130    18   66  72    76     3   1624    457
63
## -Andre Dawson       496  141    20   65  78    37    11   5628   1575
225
## -Andres Galarraga   321   87    10   39  42    30     2    396    101
12
## -Alfredo Griffin    594  169     4   74  51    35    11   4408   1133
19
##           CRuns CRBI CWalks League Division PutOuts Assists Errors
## -Andy Allanson      30   29    14      A      E      446     33     20
## -Alan Ashby         321  414   375      N      W      632     43     10
## -Alvin Davis        224  266   263      A      W      880     82     14
## -Andre Dawson       828  838   354      N      E      200     11      3
## -Andres Galarraga    48   46    33      N      E      805     40      4
## -Alfredo Griffin    501  336   194      A      W      282    421     25
##           Salary NewLeague
## -Andy Allanson      NA      A
## -Alan Ashby         475.0    N
## -Alvin Davis        480.0    A
```

```
## -Andre Dawson      500.0      N
## -Andres Galarrraga  91.5      N
## -Alfredo Griffin   750.0      A
```

1. Clean the data by removing all rows that have missing value in any variable. You will be working with this clean data.

```
hc = na.omit(hit)
```

2. Split the data randomly into train-test keeping 80% as training sample and remaining as test sample.

```
nrow(hc)

## [1] 263

set.seed(123)
trsample=sample(1:263,210)
train = hc[trsample,]
test = hc[-trsample,]
```

3. Perform the best subset selection on the training sample. While choosing the best model fit up to a 19 variable model. Choose the best model according to adjusted R squared, BIC , AIC and Cp values.

```
library(leaps)

## Warning: package 'leaps' was built under R version 4.5.2

library(stargazer)

## Warning: package 'stargazer' was built under R version 4.5.2

##
## Please cite as:

## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary
## Statistics Tables.

## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

m1=regsubsets(Salary~.,data=train,nvmax = 19)
s1 = summary(m1)

r2 = s1$adjr2
b = s1$bic
cp = s1$cp

coef(m1,which.min(b))

## (Intercept)      Walks      CAtBat      CHits      CHmRun
## DivisionW
```

```
## 149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -
129.0165012
```

```
coef(m1,which.max(r2))
```

```
## (Intercept)      AtBat      Hits      Walks      CAtBat
CRuns
## 154.1594365   -1.8722264    5.7547106    6.3852096   -0.2092154
1.7712871
##      CRBI      CWalks   DivisionW      PutOuts      Assists
Errors
##  1.1419024   -0.8989348 -121.8633686    0.2013298    0.5363804   -
6.4726568
##   NewLeagueN
##    75.8378982
```

```
coef(m1,which.max(cp))
```

```
## (Intercept)      CRuns
## 198.4540472    0.9668301
```

The best model under Best Subset Selection method with BIC as the criterion is :

Salary = 149.5853608 + 5.9582020 Walks - 0.5241611 CAtBat + 2.0687418 CHits + 1.7652811 CHmRun - 129.0165012 Division(West)

The best model under Best Subset Selection method with Adjusted R² as the criterion is :

Salary = 154.1594365 - 1.8722264 AtBat + 5.7547106 Hits + 6.3852096 Walks - 0.2092154 CAtBat + 1.7712871 CRuns + 1.1419024 CRBI - 0.8989348 CWalks - 121.8633686 Division(West) + 0.2013298 PutOuts + 0.5363804 Assists - 6.4726568 Errors + 75.8378982 NewLeague(North)

The best model under Best Subset Selection method with Cp as the criterion is :

Salary = 198.4540472 + 0.9668301 CRuns

4. Find the training MSE and test MSE for each of the best models selected according to different criterion as mentioned in 3 and compare their values.

```
# FOR BIC
```

```
mlb = lm (Salary ~ Walks + CAtBat + CHits + CHmRun + Division, data = hc)
tprb = predict(mlb, newdata = train)
tmseb = mean((train$Salary-tprb)^2,na.rm = TRUE)
tmseb
```

```
## [1] 99834.7
```

```
tepb = predict(mlb, newdata = test)
temseb = mean((test$Salary-tepb)^2,na.rm = TRUE)
temseb
```

```
## [1] 125936.5

# FOR R^2

mlr = lm (Salary ~ AtBat + Hits + Walks + CAtBat + CRuns + CRBI + CWalks +
Division + PutOuts + Assists + Errors + NewLeague, data = hc)
tprr = predict(mlr, newdata = train)
tmse1r = mean((train$Salary-tprr)^2, na.rm = TRUE)
tmse1r

## [1] 92725.29

tepr = predict(mlr, newdata = test)
temser = mean((test$Salary-tepr)^2, na.rm = TRUE)
temser

## [1] 92883.5

#FOR CP MEASURES

mlc = lm (Salary ~ CRuns, data = hc)
tprc = predict(mlc, newdata = train)
tmse1c = mean((train$Salary-tprc)^2, na.rm = TRUE)
tmse1c

## [1] 137357.4

tepc = predict(mlc, newdata = test)
temsec = mean((test$Salary-tepc)^2, na.rm = TRUE)
temsec

## [1] 143262.4
```

For BIC model, Training MSE is 99834.7 Test MSE is 125936.5

For Adjusted R² model, Training MSE is 92725.29 Test MSE is 92883.5

For Cp model, Training MSE is 137357.4 Test MSE is 143262.4

5. Repeat 3-4 using forward stepwise selection method.

```
m2=regsubsets(Salary~., data=train, nvmax = 19, method = "forward")
s2 = summary(m2)
s2

## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = train, nvmax = 19, method =
"forward")
## 19 Variables (and intercept)
##              Forced in Forced out
## AtBat          FALSE          FALSE
## Hits           FALSE          FALSE
```

```

## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: forward
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns
CRBI
## 1 ( 1 ) " " " " " " " " " " " " " " " " " " " " " "
" "
## 2 ( 1 ) " " " " " " " " " " " " " " " " " " " " "
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## 12 ( 1 ) " " " " " " " " " " " " " " " " " " " " "
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## 13 ( 1 ) "*" " " " " " " " " " " " " " " " " " " " "
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## 14 ( 1 ) "*" "*" " " " " " " " " " " " " " " " " " "
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```

```

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## 19 ( 1 ) "*" "*" "*" "*" "*" "*" "*" "*" "*" "*" "*" "*"
"*"

```

```

##          CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1 ( 1 ) " " " " " " " " " " " "
## 2 ( 1 ) " " " " " " " " " " " "
## 3 ( 1 ) "*" " " " " " " " " " " " "
## 4 ( 1 ) "*" " " "*" " " " " " " " "
## 5 ( 1 ) "*" " " "*" " " " " " " " "
## 6 ( 1 ) "*" " " "*" " " " " " " " "
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## 8 ( 1 ) "*" " " "*" " " " " " " " "
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## 10 ( 1 ) "*" " " "*" " " " " " " "*"
## 11 ( 1 ) "*" " " "*" " " " " " " "*"
## 12 ( 1 ) "*" " " "*" " " " " " " "*"
## 13 ( 1 ) "*" " " "*" " " " " " " "*"
## 14 ( 1 ) "*" " " "*" " " " " " " "*"
## 15 ( 1 ) "*" " " "*" " " " " " " "*"
## 16 ( 1 ) "*" " " "*" " " " " " " "*"
## 17 ( 1 ) "*" " " "*" " " " " " " "*"
## 18 ( 1 ) "*" " " "*" " " " " " " "*"
## 19 ( 1 ) "*" "*" "*" " " " " " " "*"

```

```
r22 = s1$adjr2
```

```
b2 = s1$bic
```

```
cp2 = s1$cp
```

```
coef(m2,which.min(b2))
```

```

## (Intercept)          Walks          CRuns          CRBI          CWalks
DivisionW
## 36.5684319      8.0759181      0.6573918      0.7699755     -0.9132118 -
150.6361441

```

```
coef(m2,which.max(r22))
```

```

## (Intercept)          Walks          Years          CAtBat          CHits
CRuns
## 142.2438912      5.0866871     -9.3715553     -0.4123632      1.0375990
1.2264552
##          CRBI          CWalks          DivisionW          PutOuts          Assists
Errors
## 0.9166203     -0.4982937    -126.5815395      0.1573527      0.4433298 -
7.9631878

```

```
## NewLeagueN
## 73.6856084

coef(m2, which.max(cp2))

## (Intercept)      CRuns
## 198.4540472    0.9668301
```

The best model under Forward Selection method with BIC as the criterion is :

Salary = 36.5684319 + 8.0759181 Walks + 0.6573918 CRuns + 0.7699755 CRBI - 0.9132118 CWalks - 150.6361441 Division(West)

The best model under Forward Selection method with Adjusted R² as the criterion is :

Salary = 142.2438912 + 5.0866871 Walks - 9.3715553 Years - 0.4123632 CAtBat + 1.0375990 CHits + 1.2264552 CRuns + 0.9166203 CRBI - 0.4982937 CWalks - 126.5815395 Division(West) + 0.1573527 PutOuts + 0.4433298 Assists - 7.9631878 Errors + 73.6856084 NewLeague(North)

The best model under Forward Selection method with Cp as the criterion is :

Salary = 198.4540472 + 0.9668301 CRuns

FOR BIC

```
mlb2 = lm (Salary ~ Walks + CRuns + CRBI + CWalks + Division, data = hc)
tprb2 = predict(mlb2, newdata = train)
tmseb2 = mean((train$Salary-tprb2)^2, na.rm = TRUE)
tmseb2
```

```
## [1] 103972.3
```

```
tepb2 = predict(mlb2, newdata = test)
temseb2 = mean((test$Salary-tepb2)^2, na.rm = TRUE)
temseb2
```

```
## [1] 121965.5
```

FOR R²

```
mlr2 = lm (Salary ~ Years + CHits + Walks + CAtBat + CRuns + CRBI + CWalks +
Division + PutOuts + Assists + Errors + NewLeague, data = hc)
tprr2 = predict(mlr2, newdata = train)
tmse1r2 = mean((train$Salary-tprr2)^2, na.rm = TRUE)
tmse1r2
```

```
## [1] 95071.25
```

```
tepr2 = predict(mlr2, newdata = test)
temse2 = mean((test$Salary-tepr2)^2, na.rm = TRUE)
temse2
```

```
## [1] 104803.7

#FOR CP MEASURES

mlc2 = lm (Salary ~ CRuns, data = hc)
tprc2 = predict(mlc2, newdata = train)
tmse1c2 = mean((train$Salary-tprc2)^2, na.rm = TRUE)
tmse1c2

## [1] 137357.4

tepc2 = predict(mlc2, newdata = test)
temsec2 = mean((test$Salary-tepc2)^2, na.rm = TRUE)
temsec2

## [1] 143262.4
```

For BIC model, Training MSE is 103972.3 Test MSE is 121965.5

For Adjusted R² model, Training MSE is 95071.25 Test MSE is 104803.7

For Cp model, Training MSE is 137357.4 Test MSE is 143262.4

6. Repeat 3-4 using backward stepwise selection method.

```
m3=regsubsets(Salary~.,data=train,nvmax = 19,method = "backward")
s3 = summary(m3)
s3

## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = train, nvmax = 19, method =
## "backward")
## 19 Variables (and intercept)
##              Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN       FALSE      FALSE
## DivisionW     FALSE      FALSE
## PutOuts       FALSE      FALSE
## Assists       FALSE      FALSE
## Errors        FALSE      FALSE
```



```

## NewLeagueN      FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: backward
##               AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns
CRBI
## 1 ( 1 ) " " " " " " " " " " " " " " " " " " " " " "
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## 18 ( 1 ) "*" "*" "*" "*" " " " " "*" "*" "*" "*" "*" "*"
"*"
## 19 ( 1 ) "*" "*" "*" "*" " " " " "*" "*" "*" "*" "*" "*"
"*"

##               CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1 ( 1 ) " " " " " " " " " " " "
## 2 ( 1 ) " " " " " " " " " " " "
## 3 ( 1 ) " " " " " " " " " " " "
## 4 ( 1 ) " " " " " " " " " " " "
## 5 ( 1 ) " " " " "*" " " " " " " "
## 6 ( 1 ) " " " " "*" "*" " " " " " "

```

```
## 7 ( 1 ) " " " " "*" "*" " " " " " "
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## 16 ( 1 ) "*" " " "*" "*" "*" "*" "*" "*"
## 17 ( 1 ) "*" " " "*" "*" "*" "*" "*" "*"
## 18 ( 1 ) "*" " " "*" "*" "*" "*" "*" "*"
## 19 ( 1 ) "*" "*" "*" "*" "*" "*" "*" "
```

```
r23 = s1$adjr2
b3 = s1$bic
cp3 = s1$cp
```

```
coef(m3,which.min(b3))
```

```
## (Intercept)          Walks          CAtBat          CHits          CHmRun
DivisionW
## 149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -
129.0165012
```

```
coef(m3,which.max(r23))
```

```
## (Intercept)          AtBat          Hits          Walks          CAtBat
CHits
## 128.4775362   -1.6435471    4.8956735    6.2891408   -0.3333956
1.0605358
##          CHmRun          CRuns          CWalks    DivisionW          PutOuts
Assists
##    2.2338055    0.9923502   -0.6084893  -119.7833172    0.1891262
0.3331968
##    NewLeagueN
##    67.0051884
```

```
coef(m3,which.max(cp3))
```

```
## (Intercept)          CHits
## 206.1775662    0.4746584
```

The best model under Backward Selection method with BIC as the criterion is :

Salary = 149.5853608 + 5.9582020 Walks - 0.5241611 CAtBat + 2.0687418 CHits + 1.7652811 CHmRun - 129.0165012 Division(West)

The best model under Backward Selection method with Adjusted R² as the criterion is :

Salary = 128.4775362 - 1.6435471 AtBat + 4.8956735 Hits + 6.2891408 Walks - 0.3333956
CAtBat + 1.0605358 CHits + 2.2338055 CHmRun + 0.9923502 CRuns - 0.6084893 CWalks -
119.7833172 Division(West) + 0.1891262 PutOuts + 0.3331968 Assists + 67.0051884
NewLeague(North)

The best model under Backward Selection method with Cp as the criterion is :

Salary = 206.1775662 + 0.4746584 CHits

FOR BIC

```
mlb3 = lm (Salary ~ Walks + CAtBat + CHits + CHmRun + Division, data = hc)
tprb3 = predict(mlb3, newdata = train)
tmseb3 = mean((train$Salary-tprb3)^2, na.rm = TRUE)
tmseb3
```

```
## [1] 99834.7
```

```
tepb3 = predict(mlb3, newdata = test)
temseb3 = mean((test$Salary-tepb3)^2, na.rm = TRUE)
temseb3
```

```
## [1] 125936.5
```

FOR R^2

```
mlr3 = lm (Salary ~ CHits + Walks + CAtBat + CRuns + AtBat + CWalks +  
Division + PutOuts + Assists + NewLeague + Hits + CHmRun, data = hc)
tprr3 = predict(mlr3, newdata = train)
tmse1r3 = mean((train$Salary-tprr3)^2, na.rm = TRUE)
tmse1r3
```

```
## [1] 93695.94
```

```
tepr3 = predict(mlr3, newdata = test)
temser3 = mean((test$Salary-tepr3)^2, na.rm = TRUE)
temser3
```

```
## [1] 91170.24
```

#FOR CP MEASURES

```
mlc3 = lm (Salary ~ CHits, data = hc)
tprc3 = predict(mlc3, newdata = train)
tmse1c3 = mean((train$Salary-tprc3)^2, na.rm = TRUE)
tmse1c3
```

```
## [1] 143058.5
```

```
tepc3 = predict(mlc3, newdata = test)
temsec3 = mean((test$Salary-tepc3)^2, na.rm = TRUE)
temsec3
```

```
## [1] 136069.5
```

For BIC model, Training MSE is 99834.7 Test MSE is 125936.5

For Adjusted R^2 model, Training MSE is 93695.94 Test MSE is 91170.24

For Cp model, Training MSE is 143058.5 Test MSE is 136069.5

7. Compare the results of the three methods in terms of choice of best model, train MSE and test MSE.

The Cp criterion produced very simple models with only one predictor in most cases. These models had the highest training and test MSE values, indicating underfitting.

The Models selected using Adjusted R squared included much more predictors and showed better predictive performance.

Among all models, the Best Subset Selection model chosen using Adjusted R squared had the lowest training MSE. The Backward Selection model chosen using Adjusted R squared had the lowest test MSE of 91170.24. The BIC models were more than the Adjusted R squared models but had slightly higher test error.

So, in Conclusion,

Best Subset Selection examines all possible combinations of predictors and chooses the optimal model, but it is computationally expensive. Forward Selection builds the model by adding predictors one at a time, while Backward Selection begins with the full model and removes predictors step by step.

Based on the test MSE values, the Backward Selection model selected using Adjusted R squared performs the best for predicting baseball players' salaries.

PROBLEM SHEET 7: Shrinkage Methods

Comparison of least squares, ridge regression, LASSO and Elastic Net

You would be working with Hitters data set available in the ISLR library. You have to predict the Salary based on all available variables. First make sure that you have deleted all rows with one or more missing values.

```
hclean = na.omit(Hitters)
```

1. Split the data set into a training set and a test set randomly keeping 100 observations in test set and remaining in the training set.

```
set.seed(123)
nrow(hclean)
```

```
## [1] 263
```

```
trsamp=sample(1:263,100)
traind = hclean[-trsamp,]
testd = hclean[trsamp,]
```

2. Train the model using least squares and report the train MSE and test MSE.

```
fit = lm(Salary ~. , data = traind)
fit2 = lm(Salary ~. , data = testd)
trainmse = mean (resid(fit)^2)
trainmse
```

```
## [1] 105334.2
```

```
testmse = mean (resid(fit2)^2)
testmse
```

```
## [1] 87901.66
```

Training MSE is 105334.2 Test MSE is 87901.66

3. Fit ridge regression models on the training set, with grid values of λ .

```
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.5.2
```

```
## Loading required package: Matrix
```

```
## Loaded glmnet 4.1-10
```

```
glmnet(traind,traind$Salary,alpha = 0)
```

```
##
```

```
## Call: glmnet(x = traind, y = traind$Salary, alpha = 0)
```

```
##
```

```
##      Df  %Dev Lambda
## 1    20  0.00 449300
## 2    20  0.73 409400
## 3    20  0.80 373000
## 4    20  0.88 339900
## 5    20  0.96 309700
## 6    20  1.06 282200
## 7    20  1.16 257100
## 8    20  1.27 234300
## 9    20  1.39 213400
## 10   20  1.52 194500
## 11   20  1.67 177200
## 12   20  1.83 161500
## 13   20  2.00 147100
## 14   20  2.19 134100
## 15   20  2.39 122100
## 16   20  2.62 111300
## 17   20  2.86 101400
```

##	18	20	3.13	92400
##	19	20	3.42	84190
##	20	20	3.73	76710
##	21	20	4.08	69890
##	22	20	4.45	63680
##	23	20	4.85	58030
##	24	20	5.29	52870
##	25	20	5.76	48180
##	26	20	6.28	43900
##	27	20	6.83	40000
##	28	20	7.42	36440
##	29	20	8.06	33210
##	30	20	8.75	30260
##	31	20	9.49	27570
##	32	20	10.28	25120
##	33	20	11.12	22890
##	34	20	12.02	20850
##	35	20	12.98	19000
##	36	20	13.99	17310
##	37	20	15.06	15780
##	38	20	16.19	14370
##	39	20	17.38	13100
##	40	20	18.63	11930
##	41	20	19.94	10870
##	42	20	21.30	9907
##	43	20	22.71	9027
##	44	20	24.17	8225
##	45	20	25.68	7495
##	46	20	27.24	6829
##	47	20	28.83	6222
##	48	20	30.46	5669
##	49	20	32.12	5166
##	50	20	33.81	4707
##	51	20	35.52	4289
##	52	20	37.26	3908
##	53	20	39.00	3560
##	54	20	40.76	3244
##	55	20	42.53	2956
##	56	20	44.30	2693
##	57	20	46.08	2454
##	58	20	47.86	2236
##	59	20	49.65	2037
##	60	20	51.44	1856
##	61	20	53.23	1692
##	62	20	55.02	1541
##	63	20	56.81	1404
##	64	20	58.60	1280
##	65	20	60.38	1166
##	66	20	62.17	1062
##	67	20	63.96	968

## 68	20	65.73	882
## 69	20	67.50	804
## 70	20	69.26	732
## 71	20	71.00	667
## 72	20	72.72	608
## 73	20	74.41	554
## 74	20	76.08	505
## 75	20	77.71	460
## 76	20	79.30	419
## 77	20	80.84	382
## 78	20	82.33	348
## 79	20	83.77	317
## 80	20	85.14	289
## 81	20	86.45	263
## 82	20	87.69	240
## 83	20	88.86	218
## 84	20	89.96	199
## 85	20	90.98	181
## 86	20	91.93	165
## 87	20	92.81	151
## 88	20	93.62	137
## 89	20	94.35	125
## 90	20	95.02	114
## 91	20	95.62	104
## 92	20	96.17	95
## 93	20	96.65	86
## 94	20	97.09	79
## 95	20	97.47	72
## 96	20	97.81	65
## 97	20	98.11	59
## 98	20	98.37	54
## 99	20	98.60	49
## 100	20	98.80	45